Assignment 1: Report

November 3, 2026

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1.1 MLP

Started with a baseline model with hidden_size = 256, dropout = 0.2, optimizer = SGD, and learning_rate = 0.01. Used a batch of 64 and 10 epochs. Here, we varied the activation layers. Tried out ReLU, LeakyReLU, and GELU. Using a 90/10 split for training and validation, LeakyReLU resulted in the highest validation accuracy ō.9018.

Will continue varying the optimizers, using LeakyReLU as the activation function. Tried out SGD, Adam, and AdamW. Again using a 90/10 split for training and validation, AdamW resulted in the highest validation accuracy ō.8650.

Tried out learning rate [0.001, 0.01, 0.1]. Learning rate ō.01 resulted in the highest validation accuracy of 0.9025.

Tried out hidden sizes of 128, 256, and 512. Hidden size of 128 was the best, with a validation accuracy of 0.8722.

Tried out dropout values of 0.0, 0.2, and 0.5. Dropout ō.0 resulted in the highest validation accuracy of 0.8788.

1.2 CNN